FLOODCAST

Predict daily river discharge

A regression model using weather and basin characteristics, with an interactive river map for exploring observations and scenarios.

Public-data experiment · November 2026

Team Straw_Hats

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Three public data sources

312,363 daily records

55 stations · 2010-01-01 to 2025-12-31

USGS discharge	Observed daily mean river flow in cfs; actual station metadata.
Open-Meteo weather	Precipitation, min/max temperature and soil moisture from reanalysis.
USGS StreamStats	Published drainage area, basin elevation, forest cover and slope where available.

Daily mean is the public experiment target; confirm the event definition of “peak daily discharge.”

Weather, basin features, and time

Features

Local weather, physical basin descriptors and hydrologic-region rainfall proxies with up to 90 days of weather history.

Models

Raw/log gradient boosting, peak weighting and regional weather proxies. Ensemble weights selected on validation only.

Validation

Hold out later dates and complete stations. Freeze ensemble selection on validation. 2025 is a reused public benchmark.

Selected model: ensemble_v2 · 293,057 training rows

Measured regression performance

Evaluation	RMSE (cfs)	Peak RMSE*	MAE (cfs)	R^2
Later dates	47,232	53,138	16,816	0.881
Unseen stations	72,109	115,119	27,946	0.901
2025 benchmark	46,543	111,777	14,003	0.915

Holdout station-median baseline RMSE: 85,500 cfs

*Extreme-event RMSE: actual discharge in the top 10% within each station. Public experiment results; not official competition scores.

Limitations and submission readiness

Flood peaks and unseen basins remain difficult

Gauge-point reanalysis does not represent every part of a large basin. Reservoir operations and incomplete basin attributes can limit accuracy.

A scenario is not a hydraulic forecast

The 3D flood overlay is an elevation-based exploration. It is separate from predicted discharge and omits channel hydraulics and flood defenses.

Before the official submission

Confirm the target and team names. Run against official test rows. Validate the generated CSV; do not edit predictions by hand.